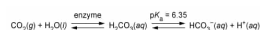


The bicarbonate (HCO_3^-) buffer system (Reaction 1) helps maintain acid-base homeostasis and a blood pH near 7.4. Therefore, concentrations of carbon dioxide (CO_2) and HCO_3^- must be tightly regulated through adaptations in respiratory and renal physiology. For patients with suspected acid-base imbalance, these concentrations can be monitored by *blood gas analysis* performed on arterial or venous blood.



Reaction 1

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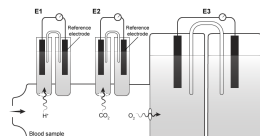


Figure 1 Automated blood gas analyzer

A pH-sensitive glass electrode (E1) is separated from the blood sample by a membrane permeable to H^+ ions. The sample pH is determined according to the voltage difference between E1 and a reference electrode maintained in a solution of known pH.

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Subsequently, using the concentration of H_2CO_3 obtained from Equation 1, the HCO_3^- concentration resulting from the decomposition of H_2CO_3 can then be calculated by applying the Henderson-Hasselbalch equation, where HA represents a weak acid and A^- represents its conjugate base:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

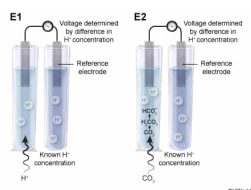
Equation 2

The reference electrode for E2 should be maintained in a solution of a known, stable concentration of:

- ✓ ☒ A. $\text{H}^+(\text{aq})$ [36%]
☐ B. $\text{HCO}_3^-(\text{aq})$ [23%]
☐ C. dissolved CO_2 [30%]
☐ D. dissolved O_2 [9%]

Correct
 37% answered correctly

Explanation:



A **reference electrode** is an electrode that has a **known electric potential** and **stable concentration** of reacting species. It is often used in combination with a glass electrode (eg, an ion-selective electrode) so that a measurable voltage (ie, potential) difference can be generated between the two electrodes.

A **pH electrode** (ie, a type of glass electrode) is sensitive to the amount of **H^+ ions** present in a solution and must be connected to a reference electrode containing a standard concentration of H^+ ions. This connection forms a concentration gradient between the pH electrode and reference electrode (ie, a **concentration cell**), which generates a voltage difference. This voltage difference is used to determine $[\text{H}^+]$ (ie, the **molar concentration** of H^+ ions present in the solution). $[\text{H}^+]$ is directly related to the **pH** of the solution by the equation:

$$\text{pH} = -\log[\text{H}^+]$$

The passage states that the CO_2 in the blood sample diffuses across a membrane to a modified pH-sensitive glass electrode (E2), where the CO_2 reacts with an aqueous solution to form H^+ ions. Therefore, the reference electrode should be maintained in a standard concentration of H^+ ions to give a known, stable value against which the concentration and voltage difference generated by the CO_2 reaction can be measured.

(Choices B and C) As stated in the passage, CO_2 reacts to produce H^+ and it is this change, not the change in CO_2 or HCO_3^- , that is measured. Therefore, the reference electrode should be maintained at a standard concentration of H^+ .

(Choice D) O_2 gas is not involved in the mechanism used at E2.

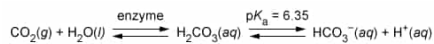
Educational objective:

A reference electrode contains a known electric potential and concentration of ions. When both electrodes in a cell contain the same chemical species (a concentration cell), the electric potential difference (voltage) between the electrodes depends only on the relative concentrations of these species.

Time spent: 0 sec
 Subject:
 General Chemistry

QID: 400239
 System:
 Solutions and Electrochemistry

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Reaction 1

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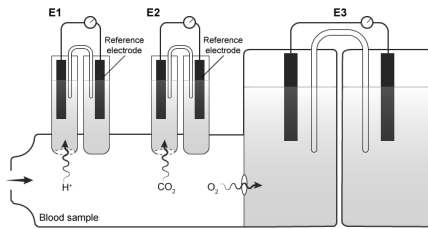


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$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Equation 2

In the Clark electrode, O_2 from the blood is:

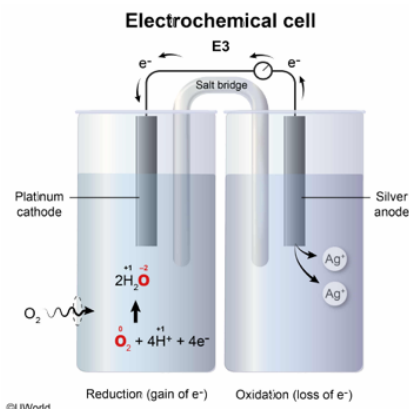
- ☐ A. oxidized at the anode. [14%]
- ☐ B. oxidized at the cathode. [2%]
- ☐ C. reduced at the anode. [8%]

✓ ☒ D. reduced at the cathode. [74%]

Correct

74% answered correctly

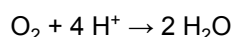
Explanation:



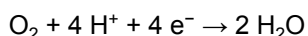
A **galvanic cell** (ie, a type of electrochemical cell) harnesses energy from a **spontaneous** (ie, energetically favorable) **oxidation-reduction** reaction by separating the reaction into half-cells. At the **anode**, one reactant is **oxidized (loses electrons)**; at the **cathode**, another reactant is **reduced (gains electrons)**. A wire connects the two half-cells and allows electrons to flow from the anode to the cathode. This produces a current that can be used to do work.

To determine whether a species is oxidized or reduced, the **oxidation state** of the element of interest must be determined. The oxidation state is the *hypothetical charge* of an atom within a molecule (ie, the charge the atom would have if it were linked to the other atoms in the molecule via ionic bonds rather than covalent bonds). In a neutral molecule (ie, a molecule with no net charge), the sum of the oxidation states of atoms must balance to zero.

At E3, dissolved O_2 gas reacts with aqueous protons (H^+) to produce water:



Initially, the oxidation state of each of the oxygen atoms in O_2 is zero because the atoms are in their elemental state. As the reaction proceeds to form H_2O , the positive charges added by the protons are balanced by the addition of two electrons (ie, 2 electrons per oxygen atom) delivered from the cathode:



Because oxygen is **more electronegative** than hydrogen, oxygen *hypothetically owns* these electrons, which gives oxygen a final oxidation state of -2 in H_2O (ie, oxygen is reduced). The hydrogen atoms (ie, the protons) each remain at an oxidation state of $+1$. Therefore, by definition this reduction reaction occurs at the cathode.

(Choice A) O_2 becomes reduced, not oxidized, when it is converted to water.

(Choice B) Oxidation occurs at the anode, not the cathode.

(Choice C) Reduction occurs at the cathode, not the anode.

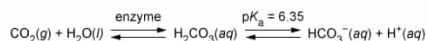
Educational objective:

Galvanic cells harness energy released by oxidation-reduction reactions. Oxidation (loss of electrons) occurs at the anode whereas reduction (gain of electrons) occurs at the cathode.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400240
System:
Solutions and Electrochemistry

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Reaction 1

In most automated blood gas analyzers, H^+ ions, CO_2 , and O_2 dissolved in the blood sample diffuse through semipermeable membranes and are measured at separate electrodes (Figure 1).

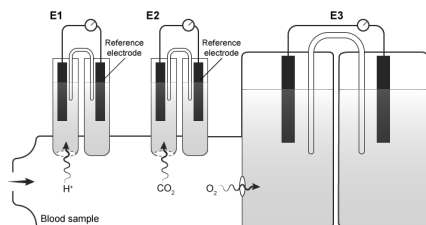


Figure 1 Automated blood gas analyzer

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$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Equation 2

Blood gas analysis results are listed below. What is the HCO_3^- concentration in the blood sample resulting from the decomposition of carbonic acid?

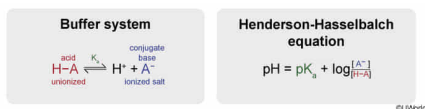
pH	7.35
P_{CO_2}	40. mmHg
P_{O_2}	90. mmHg

- ☐ A. 0.12 mmol/L [9%]
☐ B. 1.2 mmol/L [46%]
☐ C. 2.4 mmol/L [8%]
☒ D. 12 mmol/L [35%]

Correct

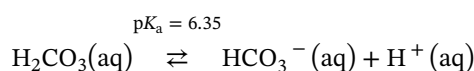
36% answered correctly

Explanation:



A **buffer system** is comprised of a mixture of a **weak acid** and its **conjugate base** or a **weak base** and its **conjugate acid**. In either type of buffer, the mixture provides both acidic and basic species to counteract added H^+ or OH^- ions and resist changes in the **pH** of the solution. A buffer has a limited pH range, known as the **buffering range**, in which it is most effective. The buffering range typically lies within **one pH** unit of the buffer's **pK_a** (ie, the pH at which half the molecules are protonated and half are deprotonated).

The **Henderson-Hasselbalch equation** (Equation 2) quantitatively describes the relationship between the pH and the pK_a of a buffer solution. For the buffer system described in the passage,



where H_2CO_3 is a weak acid (HA) and HCO_3^- is its conjugate base (A^-), the Henderson-Hasselbalch equation can be written as:

$$\text{pH} = 6.35 + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

According to the data table provided in the question, the pH of the blood sample is 7.35 and P_{CO_2} is 40. mmHg. To determine the concentration of HCO_3^- in the sample, the concentration of H_2CO_3 must first be calculated using the empirical relationship described by Equation 1:

$$[\text{H}_2\text{CO}_3] = 0.03 \times P_{\text{CO}_2} = 0.03 \times 40. = 1.2 \text{ mmol/L}$$

The HCO_3^- concentration can then be calculated by substituting the pH and the H_2CO_3 concentration into the Henderson-Hasselbalch equation:

$$7.35 = 6.35 + \log \frac{[\text{HCO}_3^-]}{1.2}$$

$$\log \frac{[\text{HCO}_3^-]}{1.2} = 1$$

Removal of the logarithm (base 10) is achieved by taking the inverse logarithm of both sides of the equation:

$$10^{\left(\log \frac{[\text{HCO}_3^-]}{1.2}\right)} = 10^1$$

$$\frac{[\text{HCO}_3^-]}{1.2} = 10$$

Solving for $[\text{HCO}_3^-]$ yields:

$$[\text{HCO}_3^-] = 12 \text{ mmol/L}$$

(Choice A) This value is obtained if the ratio of HCO_3^- to H_2CO_3 is inverted (ie, $\frac{[\text{H}_2\text{CO}_3]}{[\text{HCO}_3^-]}$ rather than $\frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$) when using the Henderson-Hasselbalch equation.

(Choice B) This value is equal to the H_2CO_3 concentration, not the HCO_3^- concentration.

(Choice C) This value corresponds to a ratio of 2:1 for the concentrations of HCO_3^- to H_2CO_3 . Because the log of $[\text{HCO}_3^-]/[\text{H}_2\text{CO}_3]$ is equal to 1, the quotient of $[\text{HCO}_3^-]/[\text{H}_2\text{CO}_3]$ is equal to 10^1 , which corresponds to an HCO_3^- concentration of 12 mmol/L.

Educational objective:

A buffer system, which consists of a mixture of a weak acid (or base) and its conjugate base (or acid), helps maintain the pH of a solution near the $\text{p}K_a$. The Henderson-Hasselbalch equation relates the pH, $\text{p}K_a$, and the concentration ratio of the acid and conjugate base.

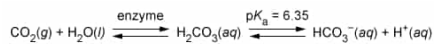
Time spent: 0 sec

Subject:
General Chemistry

QID: 400241

System:
Solutions and Electrochemistry

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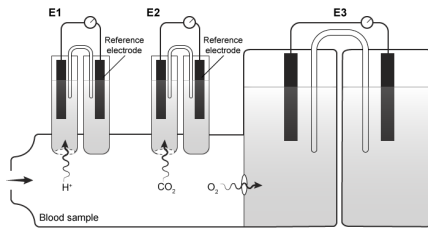


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$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Equation 2

In the alveoli, hypoventilation (decreased CO_2 gas exchange) and hyperventilation (increased CO_2 gas exchange) can lead to abnormalities in acid-base homeostasis. Which of the following is true of the *immediate* effects of abnormal ventilation?

- ☐ A. Hyperventilation decreases blood pH and decreases the HCO_3^- concentration. [11%]

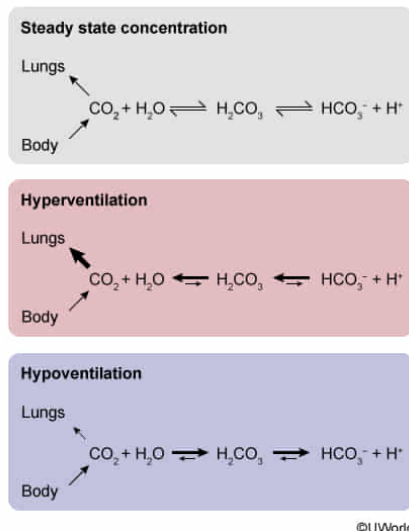
- ✓ ☒ B. Hyperventilation increases blood pH and decreases the HCO_3^- concentration. [59%]
- ☐ C. Hypoventilation increases blood pH and increases the HCO_3^- concentration. [10%]
- ☐ D. Hypoventilation decreases blood pH and decreases the HCO_3^- concentration. [17%]

Correct

59% answered correctly

Explanation:

Equilibrium: bicarbonate buffer system



According to [Le Châtelier's principle](#), chemical reactions are in a state of dynamic [equilibrium](#) to maintain optimal ratios of reactants and products. At equilibrium, the rate of the forward reaction is [equal](#) to the rate of the reverse reaction. If the equilibrium state of a system is perturbed (eg, [removal or addition](#) of a reactant or a product), the equilibrium shifts to counteract the disturbance and establish a new equilibrium. A **decrease** in the concentration of a chemical species shifts the equilibrium **toward** that species whereas an **increase** in the concentration shifts the equilibrium **away** from that species.

During ventilation, CO_2 is expelled from the lungs. An increase in ventilation (hyperventilation) decreases the concentration of CO_2 in the blood, which causes Reaction 1 to shift to the left (ie, toward CO_2) to reestablish equilibrium. This shift results in a decrease in H_2CO_3 concentration, which translates to a decrease in HCO_3^- and H^+ (ie, an increase in [pH](#)) in pulmonary circulation. Clinically, this condition is referred to as respiratory alkalosis.

(Choice A) Removal of CO_2 decreases HCO_3^- and H^+ . A decreased H^+ concentration corresponds to an *increase* in blood pH.

(Choices C and D) Hypoventilation (decreased CO_2 gas exchange) leads to an *increase* in CO_2 levels in the blood. This shifts the equilibrium to the right, resulting in increased concentrations of HCO_3^- and H^+ (ie, a decrease in pH). Clinically, this condition is referred to as respiratory acidosis.

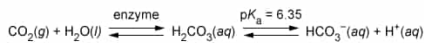
Educational objective:

Chemical reactions are in a state of dynamic equilibrium to maintain equilibrium. According to Le Châtelier's principle, changes in conditions shift the equilibrium to counteract the changes. If a species is removed, the reaction shifts toward the removed species; if a species is added, the reaction shifts away from the added species.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400242
System:
Solutions and Electrochemistry

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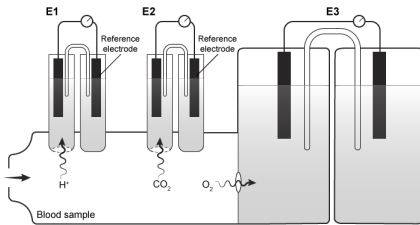


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$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Equation 2

A 0.20 mL blood sample has a P_{CO_2} of 40. mmHg. During analysis, the CO_2 in the sample diffuses across a membrane into the E2 electrode containing 0.30 mL of an aqueous solution and reacts according to Reaction 1. At equilibrium, the P_{CO_2} in the combined sample-electrode system will be:

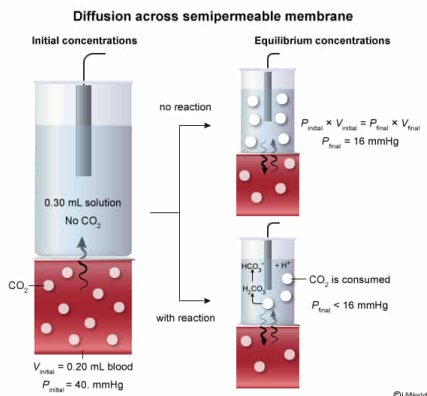
- ☐ A. equal to 0 mmHg. [4%]
☒ B. less than 16 mmHg. [45%]

- ☐ C. equal to 16 mmHg. [22%]
☐ D. greater than 16 mmHg. [26%]

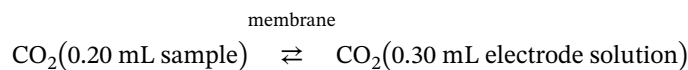
Correct

46% answered correctly

Explanation:



When a membrane permeable to an uncharged species is used to separate two solutions, the species **diffuses down its concentration gradient** (ie, moves from a higher concentration region to a lower concentration region) until the species concentration is **equal on both sides**. For instance, if 0.20 mL of a solution has a P_{CO_2} of 40. mmHg and the gas diffuses across a membrane into 0.30 mL of an electrode solution, diffusion will continue until the concentration of CO_2 is equal in both solution volumes and an equilibrium is established:



Essentially, this diffusion scenario can be viewed as a **dilution** of the CO_2 . As described in this question, all the CO_2 (

$P_{\text{CO}_2} = 40. \text{ mmHg}$) is initially confined to a 0.20 mL sample of blood. During analysis, CO_2 diffuses across the electrode membrane into 0.30 mL of electrode solution. The CO_2 is distributed between the two volumes and is free to move within a **total volume** of 0.50 mL (0.20 mL sample + 0.30 mL electrode solution). As such, the final concentration after diffusion can be calculated as:

$$\begin{aligned} P_{\text{CO}_2}(\text{initial}) \times V_{\text{initial}} &= P_{\text{CO}_2}(\text{final}) \times V_{\text{final}} \\ (40. \text{ mmHg}) \times (0.20 \text{ mL}) &= P_{\text{CO}_2}(\text{final}) \times (0.50 \text{ mL}) \\ P_{\text{CO}_2}(\text{final}) &= \frac{(40. \text{ mmHg}) \times (0.20 \text{ mL})}{0.50 \text{ mL}} = 16 \text{ mmHg} \end{aligned}$$

However, once the CO_2 diffuses from the blood sample into the electrode solution, some of the **CO_2 is consumed** as it reacts with water to form HCO_3^- and H^+ (Reaction 1). According to [Le Châtelier's principle](#), as CO_2 is **removed** from the electrode solution, more CO_2 will continue to diffuse from the blood into the aqueous electrode solution until a new equilibrium is reached. Therefore, the equilibrium concentration of CO_2 in the sample-electrode system will be **less than 16 mmHg**.

(Choice A) The diffused CO_2 will equilibrate with the aqueous solution in the electrode. It will not be completely consumed by Reaction 1, so the CO_2 concentration in the blood sample will not reach zero.

(Choices C and D) If CO_2 did not react with the electrode solution, the equilibrium concentration of the blood sample would be exactly 16 mmHg. Because some CO_2 is consumed after diffusion across the membrane, the equilibrium concentration of CO_2 will be less than 16 mmHg.

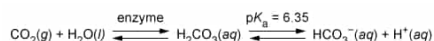
Educational objective:

An uncharged species will diffuse across a semipermeable membrane until its concentration is equal on both sides. If the species is removed from one side of the membrane, diffusion will continue in that direction until equilibrium is regained.

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General Chemistry

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The bicarbonate (HCO_3^-) buffer system (Reaction 1) helps maintain acid-base homeostasis and a blood pH near 7.4. Therefore, concentrations of carbon dioxide (CO_2) and HCO_3^- must be tightly regulated through adaptations in respiratory and renal physiology. For patients with suspected acid-base imbalance, these concentrations can be monitored by *blood gas analysis* performed on arterial or venous blood.



Reaction 1

In most automated blood gas analyzers, H^+ ions, CO_2 , and O_2 dissolved in the blood sample diffuse through semipermeable membranes and are measured at separate electrodes (Figure 1).

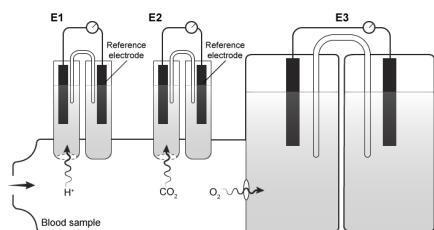


Figure 1 Automated blood gas analyzer

A pH-sensitive glass electrode (E1) is separated from the blood sample by a membrane permeable to H^+ ions. The sample pH is determined according to the voltage difference between E1 and a reference electrode maintained in a solution of known pH.

CO_2 diffuses across a gas-permeable membrane to a modified pH-sensitive glass electrode (E2), where the CO_2 reacts with water to generate HCO_3^- and H^+ ions. The partial pressure of CO_2 (P_{CO_2}) is calculated indirectly from the change in pH, as determined by the potential difference between E2 and its reference electrode.

O_2 diffuses through a second gas-permeable membrane into a Clark electrode (E3), which is an electrochemical cell consisting of a silver anode and a platinum cathode. Within the E3 cell, O_2 reacts with H^+ ions to form water. The current measured at E3 is then used to calculate the partial pressure of O_2 (P_{O_2}) in the blood sample.

The molar concentration of carbonic acid [H_2CO_3], reported in millimoles per liter (mmol/L), can be calculated from P_{CO_2} by Equation 1:

$$[\text{H}_2\text{CO}_3] \left(\frac{\text{mmol}}{\text{L}} \right) = 0.03 \times P_{\text{CO}_2} (\text{mmHg})$$

Equation 1

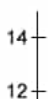
Subsequently, using the concentration of H_2CO_3 obtained from Equation 1, the HCO_3^- concentration resulting from the decomposition of H_2CO_3 can then be calculated by applying the Henderson-Hasselbalch equation, where HA represents a weak acid and A^- represents its conjugate base:

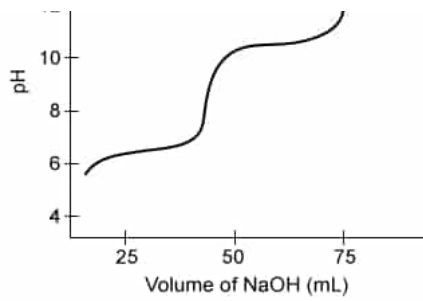
$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Equation 2

A 50 mL solution of 0.1 M carbonic acid is titrated with a solution of 0.2 M sodium hydroxide (NaOH). Which of the following represents the curve for this titration?

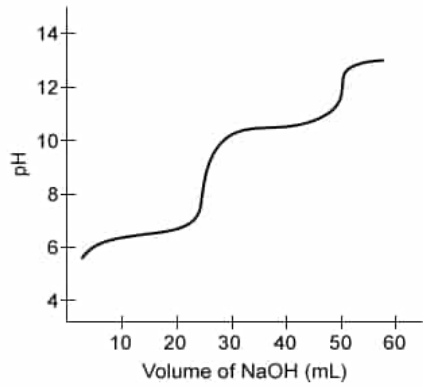
☐ A.





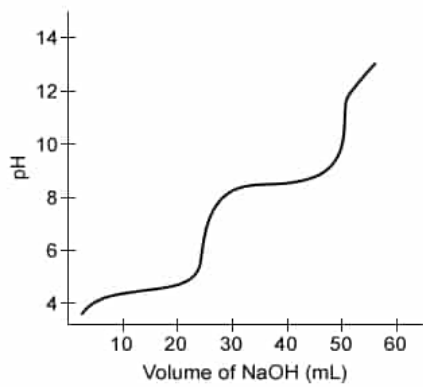
[12%]

✓ ☒ B.



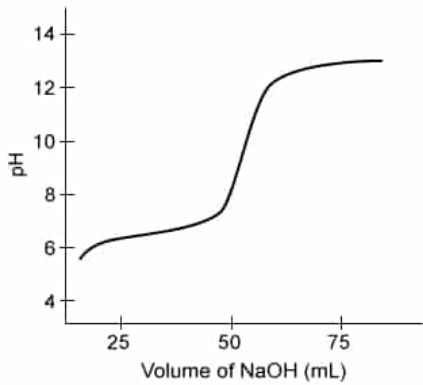
[50%]

☐ C.



[21%]

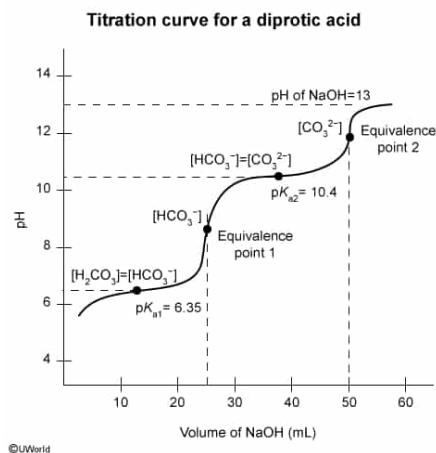
☐ D.



[15%]

Correct
50% answered correctly

Explanation:

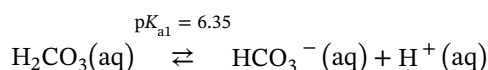


For an **acid-base titration**, the **titration curve** depicts the change in **pH** of an **acidic or basic** analyte solution (*y*-axis) as a function of the amount of base or acid titrant added (*x*-axis). The titration curve of a weak acid can be used to visualize pK_a values and equivalence points:

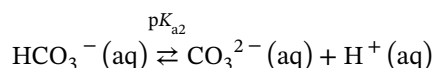
- The relatively **flat portion of the curve** corresponds to the **buffering region**. At the center of this region, the concentration of acid is equal to the concentration of conjugate base, and **$pH = pK_a$** .
- Beyond the buffering region, the **pH rises sharply**. The midpoint of this steep region is the **equivalence point**, where the moles of base are equal to the moles of the given acid species.

For a **polyprotic acid** (ie, a compound that can lose multiple acidic protons), a titration curve shows **one pK_a value** and **one equivalence point** for **each acidic proton**.

In this question, H_2CO_3 is a diprotic acid (ie, two acidic protons), and therefore has two pK_a values and two equivalence points. According to Reaction 1, the first pK_a (pK_{a1}) is 6.35 and corresponds to the equilibrium between H_2CO_3 and HCO_3^- :



The second pK_a (pK_{a2}) corresponds to the equilibrium between HCO_3^- and CO_3^{2-} :



As such, the first buffering region on the titration curve is near a pH of 6.35, and the second buffering region is at a pH greater than 6.35 (ie, $pK_{a2} > pK_{a1}$). Furthermore, the first equivalence point occurs when all 0.005 moles of H_2CO_3 are converted to HCO_3^- :

$$50 \text{ mL } H_2CO_3 \times \frac{1 \text{ L}}{1,000 \text{ mL}} \times \frac{0.1 \text{ mol } H_2CO_3}{1 \text{ L}} = 0.005 \text{ mol } H_2CO_3$$

Converting 0.005 moles of H_2CO_3 to HCO_3^- requires 0.005 moles of NaOH, which corresponds to 25 mL of a 0.2 M NaOH solution:

$$0.005 \text{ mol } H_2CO_3 \times \frac{1 \text{ mol NaOH}}{1 \text{ mol } H_2CO_3} = 0.005 \text{ mol NaOH}$$

$$0.005 \text{ mol NaOH} \times \frac{1 \text{ L NaOH}}{0.2 \text{ mol NaOH}} \times \frac{1,000 \text{ mL}}{1 \text{ L}} = 25 \text{ mL NaOH}$$

A second equivalence point occurs after another 25 mL of NaOH has been added, converting all the HCO_3^- to CO_3^{2-} . Therefore, the titration curve shown in Choice B best represents the results of the H_2CO_3 and NaOH titration.

(Choice A) This curve shows the first equivalence point occurring at 50 mL rather than at 25 mL of NaOH.

(Choice C) In this curve, pK_{a1} occurs below 6.35.

(Choice D) This curve shows the titration of a monoprotic acid with a pK_{a1} of 6.35 instead of a diprotic acid.

Educational objective:

On a titration curve for a weak acid, the pK_a corresponds to the pH at the center of the buffering zone. Beyond this zone, the sharp rise in pH corresponds to the equivalence point. For polyprotic acids, multiple pK_a values and equivalence points can be visualized.

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